

LESSON GUIDE

Technology Plus

Unit 7: Security

6.2.2 – Securing Devices

Lesson Overview:

This lesson introduces students to the methods used to secure both mobile and workstation devices. Students will understand the critical components of device security. Components include authentication, anti-malware, firewalls, software updates, and physical security mechanisms such as cable and USB locks.

Students will:

- Define and explain key security measures for devices, including authentication, anti-malware, firewalls, updates, and physical protection.
- Describe how each method contributes to overall device and data security.
- Evaluate the effectiveness of various physical security controls.

Guiding Question: What steps can be taken to secure personal and workplace devices from both digital and physical threats?

Suggested Grade Levels: 8-12

Technology Needed: Yes

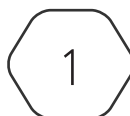
Approximate Time Required: 60-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.2 – Explain methods to secure devices and security best practices.
 - Securing devices (mobile/workstation)
 - Authentication
 - Anti-malware
 - Firewall
 - Patching/updating
 - Physical device security
 - Cable locks
 - USB locks

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Lesson Background

Background Information:

Securing devices involves both software-based and hardware-based methods. With mobile and workstation devices frequently targeted for data theft, device compromise, and network infiltration, it's critical for students to learn about layered defenses. Applying authentication strategies, keeping devices patched, installing anti-malware, and implementing physical barriers can protect both personal and professional environments.

Materials Needed:

Materials per Class	• Lesson Guide 6.2.2 – Securing Devices • Lesson Slides 6.2.2 – Securing Devices • Activity Slides 6.2.2 – Secure That Device!
Materials per Student	• Student Handout 6.2.2 – Securing Devices • Student Notes 6.2.2 – Securing Devices • Assessment 6.2.2 – Securing Devices (Optional)

Cyber Terms:

Term	Definition
authentication	the process of identifying an authorized user to provide access to a system or device
biometrics	a physical quality of a user that is used to authenticate them
multi-factor authentication	use of two or more factors to authenticate the identity of a user
anti-malware	software designed to protect devices from software-based threats like viruses or spyware by detecting and quarantining/removing them
firewall	a security tool (hardware or software) that monitors and filters network traffic to decide what data gets in or out, like a digital bouncer protecting the network
software patch	a targeted update aimed at correcting a specific issue or security flaw
cable lock	a locking mechanism used to secure a device to a structure or object
USB lock	used to block ports on a device to prevent flash drives and other devices or cables from being plugged in

6.2.2 – Securing Devices

Guiding Question: What steps can be taken to secure personal and workplace devices from both digital and physical threats?

Lesson Launch (3-5 minutes)

- Start by presenting students with the following scenario:
 - A cell phone and laptop have been left on a table in an empty room.
- Have them discuss the potential risks of leaving a device unattended and share any methods they know to keep it secure, even when no one is around.

Securing Device With Layers (5-10 minutes)

Distribute Student Handout 6.2.2 – Securing Devices to students and encourage them to fill in answers as you go through today's lesson.

Types of Devices

- Explain the differences between the types of devices with the main takeaway that **workstations** consist of larger computing devices such as laptops or desktops that have dedicated security features and applications while **mobile** devices usually include security features built into the operating system.

Layered Security

- As you progress through the security measures, reinforce that no security measure is perfect, nor will one item stop everything. It is important to implement each measure in a layered approach so should one security measure fail, another can provide backup support and defense.

Security Measures (30-35 minutes)

Authentication

- **Authentication** is often considered the first line of defense for a device or workstation as it identifies an authorized user and prevents unauthorized access.
- Proper authentication includes:
 - Using a strong password or passphrase where length is considered better than complexity, but having both is ideal.
 - **Biometrics** uses a physical quality of the user, such as facial recognition and fingerprint patterns.
 - **Multi-factor authentication (MFA)** adds an extra layer of security by requiring two or more verification steps to gain access to a device, service, or application.

Signs of Phishing

Anti-malware

- **Anti-malware** protects devices from software-based threats like viruses or spyware by detecting and quarantining/removing them.
- It is important to keep anti-malware updated as malware is constantly evolving.
- Updates include malware signatures, or unique strings of code that help identify and detect malicious software.
- Mobile devices typically have anti-malware features built into the operating system.

Firewalls

- **Firewalls** are a security tool (hardware or software) that monitors and filters network traffic to decide what data gets in or out.
- Workstations can have their own software-based firewalls while mobile devices usually do not have firewalls; however, the networks they connect to will.
- Firewalls are critical for the security of networked devices.

Updates and Patches

- **Updates** to devices typically aim to update the operating system, introducing new features, fixing errors or glitches, updating the user interface, and more.
- **Software patches** are often targeted updates aimed at correcting specific issues and in many cases involve a security vulnerability.
- As with anti-malware, maintaining updates and patches prevents your device from having security vulnerabilities and updates malware signatures.

Physical Security

- **Cable locks** prevent device theft by securing devices to desks or other physical structures or objects.
- **USB locks**, sometimes referred to as port blockers, prevent flash drives and other devices or cables from being plugged into a device.
 - This can help to prevent malicious installations through physical connections.

Activity: Secure That Device! (15-20 minutes)

- Students will work in groups to examine six scenarios that have various security issues. They will then discuss which security measures from the lesson would be best applied to help mitigate the security threat.
- Answers can be shared by groups to the whole class, or each group can be given a single scenario and then come together to discuss their findings in a jig-saw fashion.
- The scenarios:
 - The Forgotten Laptop
 - A college student leaves their laptop unattended on a table while using the restroom.

Authentication could prevent someone from using the device or using a cable lock to prevent theft.

- Questionable Link

- An employee receives an email that looks like it's from a legitimate source asking them to click a link to restore their account.

Anti-malware could prevent malicious software from being installed. Firewalls could block malicious traffic from reaching the user or from data being extracted from the device.

- Those Annoying Alerts...

- A user ignores system updates because they are busy working, and it would disrupt their work. Later the user clicks on a compromised website. Malware is unknowingly installed.

Updates or patches could close a known vulnerability. Anti-malware could prevent malicious software from being installed. Firewalls could block malicious traffic from reaching the user or from data being extracted from the device.

- US-Be Unsecure

- Someone finds a USB drive in the parking lot and plugs it into their workstation out of curiosity.

USB locks or port blockers could prevent unauthorized USB device usage. Anti-malware could prevent malicious software from being installed.

- The Conference Trap

- A user posts about an upcoming conference on social media. A personalized spear phishing email is sent with a malicious PDF attachment titled "Conference Agenda."

Anti-malware could prevent malicious software from being installed. Firewalls could block malicious traffic from reaching the user or from data being extracted from the device.

- Public Wi-Fi Mistake

- A remote worker uses a public Wi-Fi network at the airport and visits sites to interact and manage company documents while being logged in.

Authentication, specifically multi-factor authentication, could prevent an unauthorized user from accessing the company files if they were able to gain credentials from the unsecure network. Firewalls could block malicious traffic from an unknown source if the credentials have been intercepted.

Putting It All Together (3-5 minutes)

Discussion Questions

- Why is it important to use more than one security method?
Cybersecurity requires a layered approach in case one security feature fails. The failed layer is then backed up by other security measures in place.
- What risks are there with skipping or waiting an extended time to perform an update or install a patch?
Outdated or unpatched software can have security flaws or missing malware signatures making it unable to identify and therefore remove malicious threats.
- How does the digital security of a workstation compare to that of a mobile device?
Student answers will vary. Workstations usually consist of a laptop or desktop that has dedicated security programs and features that are separate from the operating systems to be used for specific threats. A mobile device relies more on the operating system to use built-in security features.